# CHANDU PRASAD BHAIRAPU



## AI Engineer

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## ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in AI solutions for digitalization and automation, from AI prototypes and agents to RAG pipelines, model evaluation and selection, LLM fine-tuning and robust Python services on Azure and AWS. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by AI consulting work with business teams and a strong foundation in machine learning.

## EXPERIENCE

### Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

**Kniit GmbH**

November 2025 – present Muldestausee, Germany

- Designed and built an AI solution end-to-end as sole developer and owned it: a knowledge-extraction system with data preparation (ingestion, chunking), prompt engineering and indexing, reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Built LLM agents and multi-agent workflows in LangGraph (entity-extraction, relationship-analysis, consolidation and reflection agents running in parallel) for complex regulatory and technical documents, and evaluated them against an SAP-system-derived ground truth test set, raising precision/recall from 33%/45% to 93%/90%.
- Built a deterministic, code-based pipeline instead of an LLM for a second, high-volume data stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing accuracy toward 100%.
- Designed a Model Context Protocol (MCP) query service that exposes the knowledge graph as a robust service with millisecond-latency retrieval, integrated into an existing tool (Claude Code) for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure where domain experts validate and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, RAG, GraphRAG, Vector Search, FastAPI, Postgres, MongoDB, Neo4j, Docker, Kubernetes, Azure, React**

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### GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

**Covestro Deutschland AG**

December 2024 – April 2025 Leverkusen, Germany

- Advised business teams on agentic AI: mapped their decision-making workflows, identified where agentic AI could support and automate steps and translated their requirements into a solution approach.
- Designed and deployed an agentic AI application with LangGraph and LangChain that routes requests across vector-database retrieval, general inference and summarization.
- Built the Python/FastAPI backend (APIs tested with Postman) and a React frontend, containerized with Docker and deployed to AWS with CDK; version control in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, Postman, React**

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### GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

**Covestro Deutschland AG**

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of RAG retrieval and generation with RAGAS and PytrecEval against chemist-provided ground truth, running as an evaluation pipeline on AWS (Lambda, SQS, DynamoDB) built with CDK for PrismaChat.
- Designed RAG pipelines (data preparation, chunking strategies, vector databases, indexing) and benchmarked multilingual embedding models on over 1M chunks of internal chemical data.
- Fine-tuned retrieval models for the RAG pipelines and trained open-source LLMs with LoRA/QLoRA fine-tuning.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search, Docker**

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### Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

**Covestro Deutschland AG**

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models on time-series process data with engineered lag and rolling-window features, and delivered a Streamlit dashboard for chemists to simulate parameter changes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, Streamlit**

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### Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

**Cognizant Technology Solutions**

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for EV load time-series forecasting; the LSTM reached 96% test accuracy with strong performance on unseen data.

## PROJECTS

**Project: Product Q&A Assistant | n8n | RAG | Python | LLM APIs**

- n8n automation workflow that sorts incoming customer emails, answers product and application questions from technical and safety data sheets with a RAG-based AI assistant (FastAPI + vector store), and routes uncertain cases to a person for approval.
- Prompts and decision logic are versioned and tested on a labelled question set; personal data is removed before any LLM call (GDPR) and answers are marked as AI-generated.

**Project: Meme Generation using LLMs | Open Llama 3B | QLoRA Fine-Tuning**

- Fine-tuned an Open Llama 3B model with QLoRA, which accelerated training, to generate responses to educational meme-related prompts.

## TECHNICAL SKILLSET

- **LLM & AI Services:** LangChain, LangGraph (agents), RAG pipelines, LLM training (LoRA/QLoRA fine-tuning), Hugging Face Transformers, LiteLLM, MCP, prompt engineering
- **ML Frameworks:** PyTorch, TensorFlow/Keras, Scikit-Learn, XGBoost; evaluation with RAGAS, PytrecEval
- **Software Engineering:** Python, FastAPI, Python microservices, REST APIs, Postman, Git, GitLab CI/CD, React (beginner)
- **Cloud & Data:** Azure, AWS (Lambda, SQS, DynamoDB, EC2, S3, CDK), Docker, Kubernetes, SQL, Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Ways of working:** stakeholder requirements and advisory, AI-assisted development with Claude Code, Agile

## EDUCATION

**Master of Science in Artificial Intelligence** — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

**Bachelor of Technology in Mechanical Engineering** — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

## PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

## LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)